

UTM TENSILE REPORT — UTM_Test_20260824_111931

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

47.1 MPa

UTS

39.4 MPa

Yield σ_y

3.17 GPa

Modulus E

2.9 %

Failure ϵ_f

1100 kJ/m³

Toughness

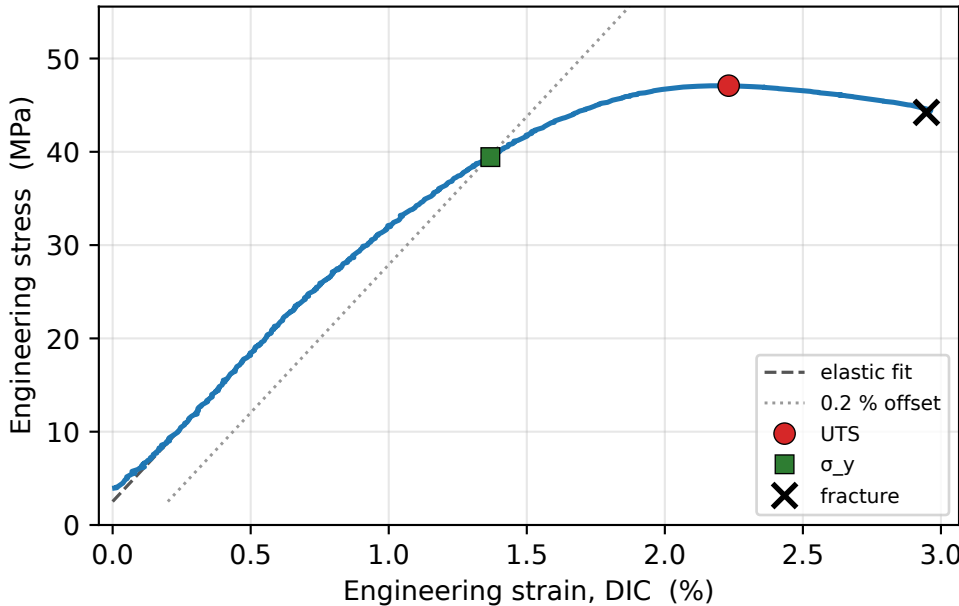
310 N

Preload anchor

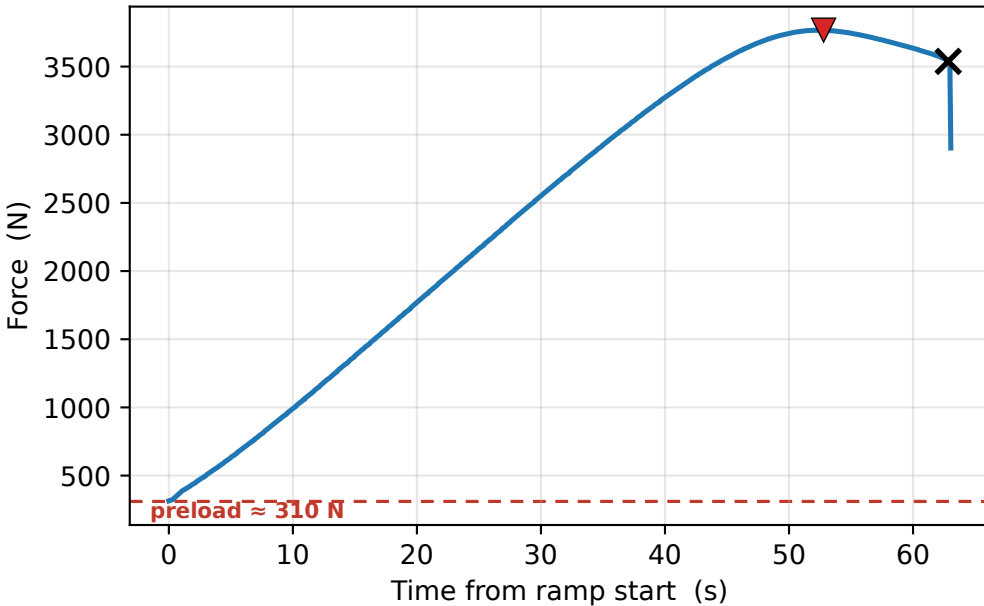
TEST SETTINGS

Specimen	UTM_Test_20260824_111931
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	99 %
Force calib	sc -0.0065 / off 334.646
Comment	—
Test date	2026-08-24 11:17:56
Duration	99.9 s

Stress-strain



Load vs time

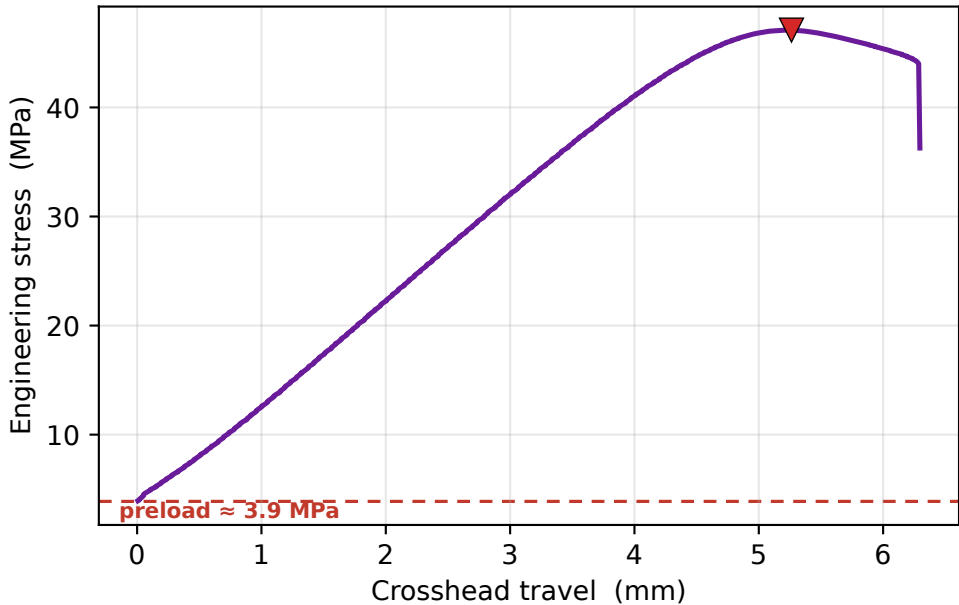


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	47.1	32-60 PASS	×1.23
σ_y	39.4	30-50 PASS	×1.47
E	3.2	3-5.5 PASS	×0.90
ϵ_f	2.9 %	(range 3-7 %)	×2.71

✓ strength inside journal range ($k \approx 1$)

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

